

TUMBES, Peru

WMO: 843700

Lat: 3.553S

Lon: 80.381W

Elev: 115

StdP: 14.63

Time Zone: -5.00 (W05)

Period: 82-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	66.8	67.8	60.3	78.4	71.5	62.3	84.4	70.7	18.6	77.8	16.4	78.5	1.8	270	0.260

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.8	89.8	78.2	89.0	77.8	88.0	77.2	79.9	86.9	79.1	86.1	78.6	85.5	9.3	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.3	147.4	83.6	77.3	142.5	82.7	76.8	140.3	82.4	43.6	87.7	42.9	85.9	42.3	85.9	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 18.0	14.9	13.0		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(5)			DB	59.2	81.9	4.7	2.8	55.8	83.9	53.1	85.5	50.4	87.1	47.0	89.1	(p)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.2	80.2	81.3	81.6	80.9	79.8	77.2	75.7	74.6	74.8	76.2	77.1	79.2
	DBStd	3.56	1.83	1.64	1.71	1.79	2.32	3.15	3.28	3.40	2.67	2.92	2.90	2.38
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10298	938	876	980	926	925	816	796	764	744	812	814	907
	CDD65	4823	473	456	515	476	460	366	331	299	294	347	364	442
	CDH74	36126	4089	4458	5145	4404	3853	2344	1762	1365	1209	1813	2202	3482
	CDH80	9485	1105	1356	1640	1346	1057	549	363	280	169	329	436	855
(14) Wind	WSAvg	4.5	4.8	4.4	4.9	4.1	3.8	3.7	4.2	4.5	4.8	5.0	4.7	5.0
(15) Precipitation	PrecAvg	19.0	2.3	5.4	6.2	3.0	0.6	0.0	0.0	0.0	0.1	0.2	0.3	0.8
	PrecMax	88.1	17.2	28.9	20.0	16.8	5.5	0.5	0.4	0.2	1.0	0.7	5.5	16.9
	PrecMin	1.6	0.0	0.0	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	17.2	3.4	6.3	5.5	3.5	1.4	0.1	0.1	0.0	0.2	0.2	1.0	3.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.6	90.7	91.1	90.3	90.0	89.3	88.9	88.7	87.2	88.0	89.2	89.7
		MCWB	77.4	78.7	79.7	78.6	78.3	77.9	77.3	76.8	76.3	76.3	76.9	77.3
	2%	DB	87.8	89.2	89.5	89.1	88.2	87.1	86.5	86.2	84.2	85.9	86.4	87.6
		MCWB	76.4	77.8	78.4	78.2	77.6	77.1	76.9	75.9	74.7	75.3	75.9	76.8
	5%	DB	86.4	87.8	88.2	87.8	86.8	85.2	83.9	82.7	81.4	83.6	84.3	86.0
		MCWB	75.8	76.9	77.6	77.5	76.7	76.1	75.3	74.2	73.1	73.7	74.2	75.8
	10%	DB	85.6	86.4	87.0	86.3	85.4	82.9	81.1	79.9	79.4	81.3	82.3	84.4
		MCWB	75.4	76.3	77.0	76.7	75.9	74.8	73.5	72.6	72.0	72.8	73.3	74.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.7	80.9	81.0	80.5	80.1	79.2	78.8	77.4	76.9	77.2	78.0	78.7
		MCDB	85.6	87.4	88.7	88.1	86.8	85.9	86.2	86.9	86.0	86.7	87.4	86.9
	2%	WB	78.7	79.4	79.7	79.2	79.0	78.1	77.3	76.2	75.3	75.8	76.5	77.6
		MCDB	85.0	86.2	86.8	86.8	85.5	84.8	85.1	85.2	82.8	84.2	85.2	85.4
	5%	WB	77.8	78.7	78.9	78.5	78.1	77.1	76.0	74.7	74.0	74.5	75.3	76.9
		MCDB	83.7	85.6	85.5	85.9	84.6	83.9	82.9	82.0	80.2	82.0	82.6	84.3
	10%	WB	76.9	77.8	78.1	77.6	77.0	75.8	74.5	73.3	72.7	73.4	74.1	75.9
		MCDB	82.5	84.2	84.4	84.5	83.5	81.8	80.3	79.1	78.5	80.6	81.0	82.7
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	11.8	11.8	12.3	11.8	11.0	10.8	10.4	9.7	10.1	10.4	11.0
		MCDBR	12.6	13.1	13.1	13.8	13.3	13.3	13.6	14.1	12.1	12.3	12.5	12.2
		MCWBR	5.6	5.4	5.5	5.6	5.5	5.9	6.7	6.9	5.9	5.8	6.0	5.5
	5% WB	MCDBR	11.7	12.6	12.5	13.4	12.9	13.2	13.6	13.7	11.9	11.5	11.8	11.8
		MCWBR	5.6	6.0	5.9	5.7	6.0	5.9	6.8	7.1	6.1	5.7	6.2	5.6
(40) Clear-Sky Solar Irradiance	taub		0.466	0.458	0.469	0.484	0.479	0.457	0.461	0.478	0.523	0.502	0.496	0.505
	taud		2.315	2.347	2.313	2.260	2.275	2.347	2.334	2.270	2.123	2.195	2.200	2.176
	Ebn at Noon		277	279	273	262	255	256	257	260	255	264	266	264
	Edh at Noon		43	42	43	44	42	38	39	43	51	49	48	49
(44) All-Sky Solar Radiation	RadAvg		1603	1577	1681	1699	1543	1406	1389	1482	1596	1650	1709	1686
	RadStd		131	89	109	68	87	86	102	97	97	80	93	97

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page